



Interactive Electrical Wiring Diagrams (iEWD)

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BEFORE USING THIS PUBLICATION

HEALTH AND SAFETY

Always follow health and safety guidelines, specifically the Electrical Precautions detailed in the Workshop Manual.

Important

The information provided in this publication is for use only by competent, qualified auto-electricians. Good product knowledge is assumed, as well as the ability to access and use recommended test equipment and other reference material provided.

Test equipment and other reference material

The information in this publication should be used in conjunction with the recommended test equipment; refer to Workshop Manual. Other reference material includes: Electrical Revision Documents (ERD), Technical Service Bulletins (TSB) and the Workshop Manual.

Battery disconnection and reconnection

It is imperative that any information relating to battery disconnection and reconnection is followed; refer to the appropriate sections in the Workshop Manual.

Fault diagnosis

Always use the recommended test equipment and test procedures for correct and reliable fault diagnosis; refer to the appropriate sections in the Workshop Manual.

NOTE: Before starting electrical checks on the vehicle, ensure that relevant mechanical functions operate satisfactorily; refer to the appropriate section of the Workshop Manual.

INTERACTIVE WIRING DIAGRAMS (iEWD)

Using The Wiring Diagrams

Introduction

The interactive wiring diagrams (iEWD) combines two separate publications, the Electrical Wiring Diagrams (EWD) and the Electrical Reference Library (ERL). The iEWD is part of the JLR diagnostic toolset, providing a new method for easily linking between the two sets of data and provides quicker:

- Navigation around the diagrams
- Identification of connector location and pinout
- Connector and data search capability

Note: The connector and data search feature is not fully developed and currently only functions for the page being viewed. In the future, a fully developed iEWD will render data that is fully searchable.

Section numbering

The sections in this publication are ordered to match the Global Outline numbering system as found in the current Workshop Manual. The Power and Ground distribution circuits can be found under section 414-01, BATTERY, MOUNTING AND CABLES.

Note: Where circuit diagrams show more than one sub-system, the circuit will be located in the section that carries the first-named sub-system, for example: Starting and Charging will be located under section 303-06 Starting System, since 'Starting' is the first-named sub-system.

Navigating the diagrams

The navigation control tool comprises three buttons: zoom in, zoom out, zoom reset.

An alternative method of navigating is:

- *Pan*

Left-click and drag

- *Zoom*

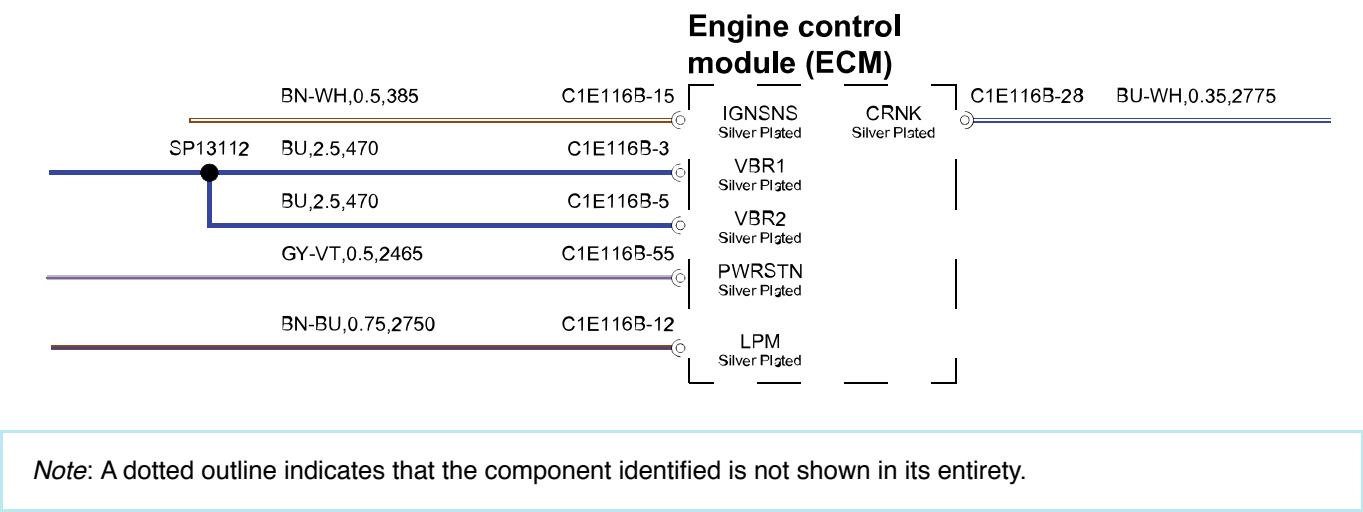
Double left-click (zoom in), Double right-click (zoom out)

To follow a wire path to another sheet from the chosen arrow head:

- Make a note of the letter positioned directly above the arrow head
- Left-click on sheet number directly below the arrow head

On the newly located sheet, find the corresponding arrowhead and letter.

Components



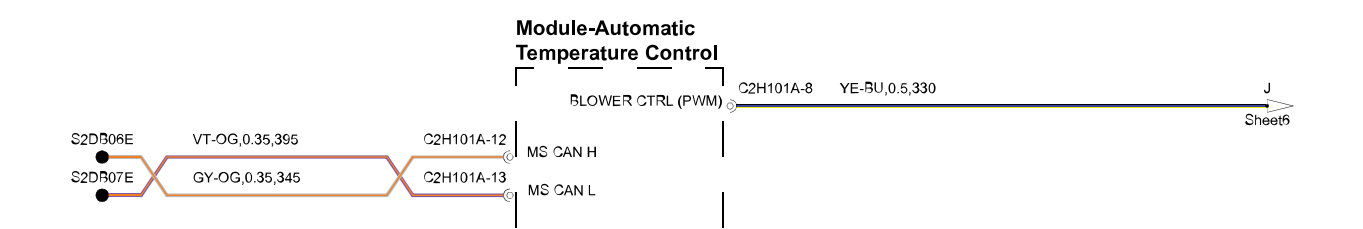
Connectors

Connectors and header joints are identified by their corresponding connector number with a numbered suffix to indicate the pin-out detail of the wire, i.e. C1E116B-15 identifies connector 1E116B, pin number 15. Wire insulation colors are listed in a table at the end of this section. Where wires have a predominant color with a secondary color tracer, the main color is identified first, i.e. WH-BK - white with a black tracer.

Wire length

The wire length (in millimetres) is displayed after the color and cross sectional area; for example, BU,2.5,470. In this example, the figure, 470 indicates the approximate position of the harness splice is 470mm from connector C1E116B.

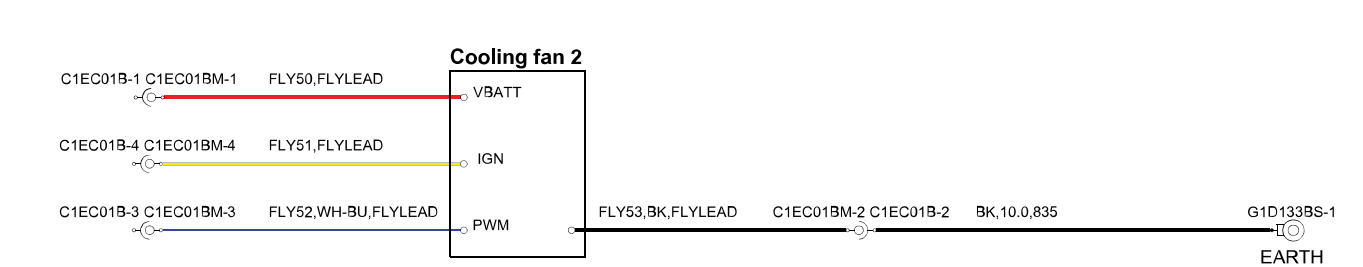
Line Types



Crossed wires as illustrated above show an example of how a twisted pair of wires may be represented on the circuits.

The arrows illustrated above show an example of the page break symbols, identifying that the circuit continues at the corresponding letter on the sheet number indicated.

Ground points



Ground points are identified with an eyelet symbol and a connector number, except where components are grounded through its fixings, when only the eyelet is shown.

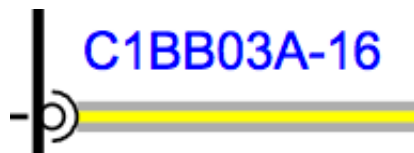
The cup and ball symbol represents the male and female halves of the connector. Most connectors plug directly into a component but some are wired directly to the component using a 'flylead' as with C1EC01BM above.

Electrical Reference Library (ERL)

Introduction

The Electrical Reference Library (ERL) data is now accessible directly by following the hyperlinks from the interactive wiring diagrams.

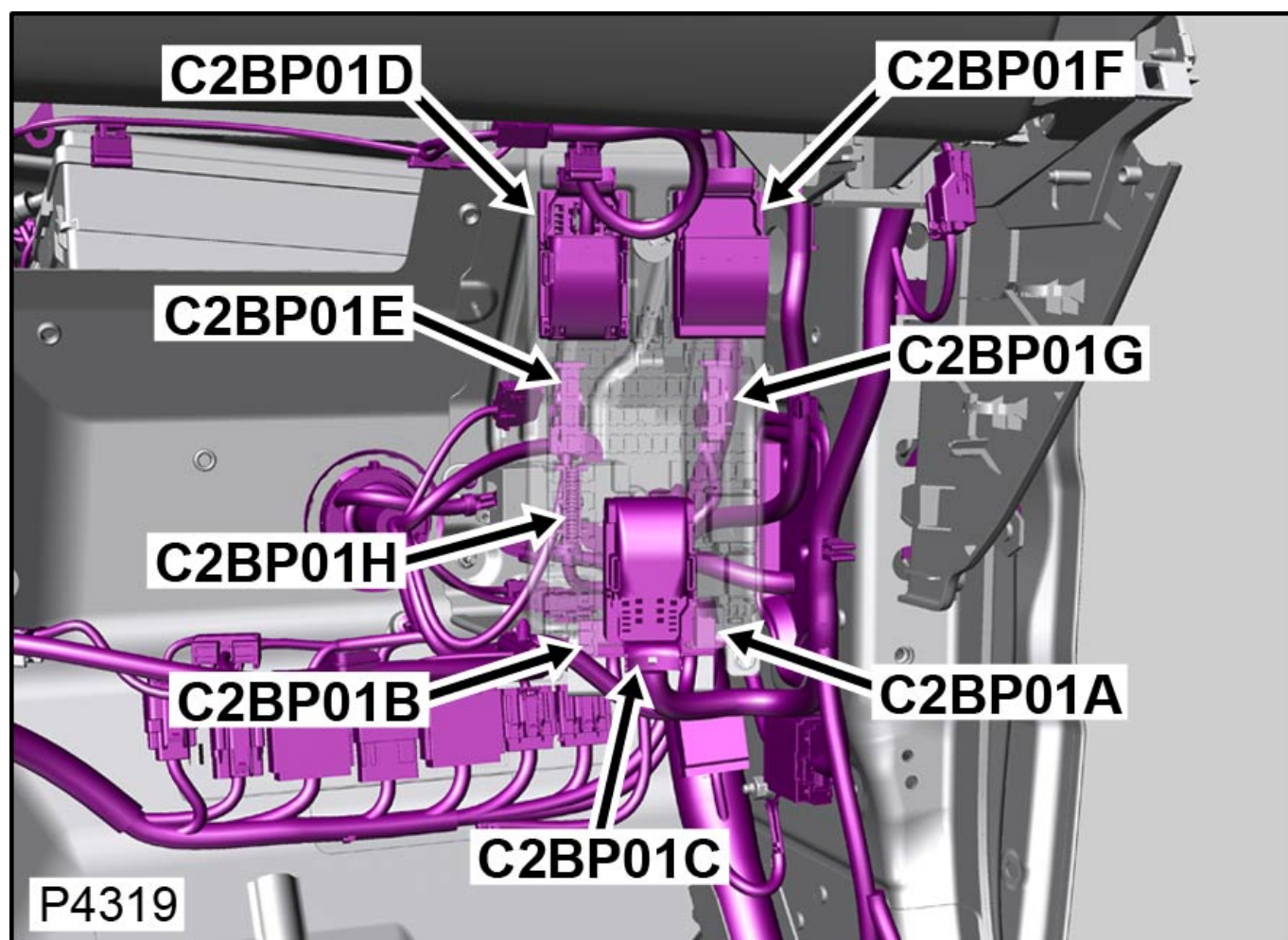
Identification of connector location and pinout



To access connector details, hover the pointer over the connector number on the wiring diagram and left-click; the details will be displayed including, where available, a graphic of the connector face-view and a location image.

The list of Connectors to the right of the location image presents the opportunity to view connector details for each annotated connector shown on that image.

Note: Connectors that are shown in the image are not necessarily part of the same electrical system, but are physically located in the area illustrated.



Refer to Connector details for an explanation of how to interpret the rest of the data.

Connector details

Connector details comprises:

- **Connector Designation** – The designated reference, prefixed 'C' or 'G' in the case of some ground connections.
- *CONNECTOR HOUSING SERVICEABLE* – The part identified is available through the normal parts ordering process.
- *CONNECTOR HOUSING NOT SERVICEABLE* – The part identified is not available.
- **Description** – Usually derived from the component to which the connection is made.
- **Location** – Used in conjunction with the photograph to indicate the location of the connector.

Note: References to the LH or RH side given in this document are made when viewing the vehicle from the rear.

- **Qualifier** – For some connectors it is not possible to represent all possible conditions. An image may show a left-hand condition, the right-hand condition may be similar and is identified as such using a Qualifier, for example: LHD condition illustrated.
- **Location Image** – Shows the location of the subject connector. For convenience, some images identify more than one connector.

Note: Images do not necessarily represent the optimum method for accessing a connector, always refer to the appropriate Service Repair Operation within the Workshop Manual before attempting to access a connector.

- **Face View** – An outline of the connector housing, viewed from the front, showing pin numbers (if applicable).
- **Part No.** – The part number as it appears on the harness drawing (if applicable).
- **Color** – If applicable, the color of the connector housing is shown. NATURAL is used to describe connectors with a clear/translucent plastic finish.
- **Cavities** – Identifies the number of cavities within the connector.
- **Harness** – Identifies the harness description (English only).
- **Wire Chart (Pin-out table)** – A five column table, giving details of each wire within the connector:

Sample Wire Chart and Service Repair Information

The five columns provides specific wiring information related to each connector.

1. **Cav**: The connector pin (cavity) number.

NOTE: Wires may not be fitted to all cavities.

2. **CSA**: The cross sectional area of the wire in mm².
3. **Col**: The color of wire populating the connector pin.

4. **Cct:** Identifies the model or option which uses the wire. 'ALL' means applicable to all models in the range fitted with the feature or system in question. In instances where different models, features or systems require different color wires to be fitted in a cavity, each instance of the cavity is included in the pin-out table.
5. **Destn:** The destination of the wire.

The last seven columns provide Service Repair Information where applicable.

NOTE: When the seven columns are not shown, no SRI is available for that particular connector.

If a connector uses more than one type of terminal (pin), different service repair information may be required.

1. **Pre-Terminated Lead:** Pre-terminated leads (including bespoke harnesses where applicable).
2. **Extract Tool:** Connector pin extractor tools.
3. **Crimpers:** Terminal applicator tool.
4. **Wire Stripper:** Wire cutter/stripper.
5. **Heat Shrink Sleeve:** Adhesive-lined heat-shrink sleeving.
6. **Ident Sleeve:** Wire color identification sleeves.
7. **Splice:** Splice connectors.

Wire chart and service repair information

Cav	CSA	Col	Cct	Destn	Pre-Terminated Lead	Extract Tool	Crimpers	Wire Stripper	Heat Shrink Sleeve	Ident Sleeve	Splice
1	2.5	GN-RD	B2CAF	C2BP01F	418-549-11	418-612	418-116A	418-117	418-104	418-112	418-107
2	1.5	GN	B2CAF	C2BP01F	-	-	-	-	-	-	-

Wire chart only

Cav	CSA	Col	Cct	Destn
1	2.5	GN-RD	B2CAF	C2BP01F
2	1.5	GN	B2CAF	C2BP01F

Abbreviations and acronyms

Abbreviation	Description
ABS	Anti-lock braking system
AUTO	Automatic transmission
AVI	Audio video input

BBUS	Battery backed-up sounder
BJB	Battery junction box
BSM	Blind spot monitoring
CAN	Controller area network
CJB	Central junction box
CVD	Continuously variable damping
D4	Diesel engine - Straight-four
DAB	Digital audio broadcasting
DBJB	Dual-battery junction box
DPF	Diesel particulate filter
DSC	Dynamic stability control
DTC	Diagnostic trouble code
DV6	Diesel engine - V6
DV8	Diesel engine - V8
EJB	Engine junction box
EGR	Exhaust gas recirculation
FFBH	Fuel fired booster heater
GPS	Global positioning system
HID	High intensity discharge
HLD	High level definition
HS CAN	High speed controller area network bus
LF	Low frequency
LH	Left hand
LHD	Left hand drive
LIN	Local interconnect network

MMM	Navigation system module
MOST	Media orientated system transport
M/S	Manual/sport
MS CAN	Medium speed controller area network bus
NA	Naturally aspirated
NAS	North American specification
PDC	Park distance control
PV6	Petrol engine - V6
PV8	Petrol engine - V8
PV8NA	Naturally aspirated engine - V8
PV8SC	Super charged engine - V8
RF	Radio frequency
RH	Right hand
RHD	Right hand drive
SAI	Secondary air injection
SCL	Steering column locking
SDARS	Satellite digital audio radio service
SRS	Supplemental restraint system
TCM	Transmission control module
TMC	Traffic message channel
TPMS	Tire pressure monitoring system
TV	Television
UK	United Kingdom
USB	Universal serial bus
VICS	Vehicle information and communication system

Wire color codes

The following list contains wire color codes used on the vehicle harnesses.

Code	Color
BK or B	Black
GN or G	Green
PK or K	Pink
LG	Light green
BN or N	Brown
OG or O	Orange
VT or P	Purple
RD or R	Red
GY or S	Grey (Slate)
T	Transparent
BU or U	Blue
WH or W	White
YE or Y	Yellow

HARNESS REPAIRS

Repair Methods

CAUTION: Do not attempt to repair any part of a wiring harness unless appropriate training has been received. Refer to the Workshop Manual, Section 418-02, Wiring Harnesses, DESCRIPTION AND OPERATION.

FIBRE OPTICS

General

The fibre optic system installed uses Media Oriented Systems Transport (MOST), which is a multimedia network, optimized for automotive applications. Its design allows it to provide a low-overhead and low-cost interface for the simplest of devices, such as microphones and speakers. More intelligent devices can automatically determine the

features and functions provided by all other devices on the network and establish sophisticated control mechanisms without distracting the driver.

Under normal installation conditions the system is robust and failures should not occur. However, since the optical fibres convey data using infra-red light, it is vital the passage of light down the fibre is unobstructed. Obstruction of infra-red light can be caused by:

- Contamination of the fibre ends.
- Damage to the fibre ends.
- Bending, kinking or damaging the cable.

NOTE: Fibres damaged by kinking or exposure of the optical core due to abrasion must be replaced.

Handling

Take special care to avoid damage or contamination when handling or working in the vicinity of fibre optic cables and connectors.

NOTE: Damage or contamination includes scratches to the cable ends and pollution caused by dust, dirt, oil or moisture.

CAUTION: When handling fibre optic cables cleanliness is of paramount importance. The fibre ends should not be touched even with clean bare hands as the natural oils deposited from the skin may penetrate the fibre or may cause dirt to adhere to the fibre end.

System malfunctions and unnecessary warranty claims can be minimised by following these guidelines:

- After disconnection of any cables, carefully install appropriate dust caps to protect the mating faces of the connectors from damage or contamination.
- Avoid introducing tight bends (less than 30mm radius) or kinks into the fibre optic cable during service or repair. Tight bends or kinks could impair operation, cause immediate system failure, or future system failure.
- Avoid excessive force, strain or stress on the fibres and connectors, especially permanent stress after reinstallation.